

FICTITIOUS PRODUCT — DEMONSTRATION DATA

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TECHNICAL SPECIFICATION DATASHEET

1/16 DIN Panel-Mount Digital PID Temperature Controller with Pt100 Input and 4–20 mA Control Output

|           |                                                     |                     |                                                   |
|-----------|-----------------------------------------------------|---------------------|---------------------------------------------------|
| SKU:      | CN-DEMO-PID-PT100-RS1                               | Model:              | CN-PID-T1                                         |
| Category: | Instrumentation & Process Control / PID Controllers | Manufacturer / Rev: | ControlNet Demo Systems   Rev 1.0 (Demonstration) |

Section 1: Identification and Model Overview

The microprocessor-based industrial controller model **CN-PID-T1** (SKU code: **CN-DEMO-PID-PT100-RS1**) is a dedicated front-panel instrument for closed-loop thermal regulation. It incorporates an advanced adaptive Auto-Tuning algorithm and a continuous proportional 4–20 mA current output for modulating actuators.

| Identification Parameter | Catalog Specification                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------|
| SKU Code                 | CN-DEMO-PID-PT100-RS1                                                                                   |
| Model Designation        | CN-PID-T1                                                                                               |
| Instrument Type          | Microprocessor-Based Digital PID Temperature Controller                                                 |
| Control Algorithm        | Advanced PID with adaptive Auto-Tuning, manual override, and selectable ON/OFF mode                     |
| Typical Application      | High-precision temperature regulation in industrial ovens, extruders, chemical reactors, and autoclaves |

Section 2: Physical Mounting and Mechanical Form Factor

Mechanical housing engineered exclusively for flush mounting through instrument panels or enclosure doors.

| Mechanical Parameter    | Nominal Specification and Mounting Attributes                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Mounting Type           | Exclusive front-panel flush mounting (Panel Mount / Control Panel Door)                                |
| Standard Form Factor    | 1/16 DIN international dimensional standard (48 mm × 48 mm front bezel)                                |
| Panel Cutout Dimensions | 45.0 mm (+0.5/-0) × 45.0 mm (+0.5/-0); accommodates panel plate thickness from 1.0 to 8.0 mm           |
| Securing Mechanism      | Rear detachable plastic mounting bracket collar with dual front-tightening tension screws              |
| Dimensions & Sealing    | 48 × 48 × 95 mm (depth behind panel: 86 mm) / IP65 front bezel sealing with included synthetic gasket  |
| Structural Restriction  | NOT SUITABLE FOR DIN RAIL MOUNTING. The unit has no base clips or mounting adapter for 35 mm DIN rails |

Section 3: Electrical Power Supply & Operational Tolerances

Internal universal switched-mode power supply with EMI filtering designed for harsh industrial electromagnetic environments.

| Electrical Parameter        | Operational Requirement and Tolerance Range                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------|
| Nominal Supply Voltage      | 100–240 VAC (50/60 Hz) universal AC line voltage                                               |
| Operating Voltage Tolerance | 85 VAC to 264 VAC steady-state without functional degradation                                  |
| Power Consumption           | 6.5 VA maximum at 240 VAC with 4–20 mA current loop under 100% full load                       |
| Dielectric Isolation        | 2000 VAC (50/60 Hz) for 1 minute between power line terminals and low-voltage signal terminals |
| Rear Terminal Connections   | 12-point rear screw barrier terminal block with M3 screws and insulating phase partitions      |

Section 4: Analog & Discrete I/O Interfaces and Sensors

Critical transducer interface configuration. The process variable input accepts exclusively 3-wire Pt100 RTDs, while the 4–20 mA interface functions strictly as a control OUTPUT.

| Channel / Signal         | Direction & Signal Type             | Detailed Technical Specification                                                                                                                                                        |
|--------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temperature Sensor Input | SINGLE INPUT (Process Variable PV)  | 3-wire Pt100 RTD input (IEC 60751); automatic lead-wire resistance compensation up to 20 Ω/wire; range: -200.0 °C to +600.0 °C; resolution: 0.1 °C; 16-bit ADC; accuracy: ±0.2% of span |
| Current Input 4–20 mA    | NONE (0 mA Inputs)                  | THE INSTRUMENT HAS NO 4–20 mA INPUT. Output ≠ Input. Does not accept analog current signals from external pressure or flow transmitters                                                 |
| Analog Control Output    | ACTIVE OUTPUT (Manipulated Var. MV) | 4–20 mA OUTPUT: Proportional active current output for modulating PID control; 14-bit DAC; max loop load 500 Ω; internally powered active loop; analog update cycle 200 ms              |
| Auxiliary Alarm Output   | Discrete Output (Relay)             | 1 × SPST electromechanical relay output (250 VAC / 3 A resistive), user-configurable for high/low temperature process alarms                                                            |

Section 5: Communications, Fieldbus and Protocol Specifications

Serial fieldbus connectivity for distributed supervisory control in plant SCADA networks.

| Fieldbus Parameter      | Supported Configuration and Protocol Details                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| Physical Serial Port    | 1 × RS-485 half-duplex on rear terminals (TRX+, TRX-, and isolated SG), with 1000 V galvanic isolation          |
| Industrial Protocol     | Modbus RTU over RS-485 in SLAVE mode. Slave node address configurable from front keypad between 1 and 247       |
| Transmission Baud Rates | 4800, 9600, 19200, and 38400 bps selectable via front configuration menu (default: 9600 bps, 8-N-1)             |
| Monitored Registers     | Holding Registers (PV: Process Value, SP: Setpoint, MV: Control Output %, Kp, Ti, Td, Alarm status)             |
| Ethernet Support        | No native Ethernet interface; connection to Modbus TCP networks requires an external serial-to-Ethernet gateway |

Section 6: Engineering Constraints and Design Contraindications

Technical incompatibility criteria and mandatory engineering design rules:

| Critical Constraint / Contraindication                                   | Technical Rationale and Design Exclusion Rule                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRITICAL CONTRAINDICATION 1:<br>PANEL != DIN (Panel Mount ≠ DIN Rail)    | The CN-PID-T1 is engineered <b>strictly for panel mounting (Panel != DIN / Panel distinct from DIN)</b> . It is strictly contraindicated to specify or attempt mounting this device on a 35 mm DIN rail. It possesses no DIN rail mounting clips or certified enclosure chassis.                                                                                                                            |
| CRITICAL CONTRAINDICATION 2:<br>OUTPUT != INPUT (4–20 mA Output ≠ Input) | The 4–20 mA terminal on this instrument is strictly a <b>4–20 mA OUTPUT</b> for proportional control (MV). <b>Output != Input (Output is not a 4–20 mA input): This instrument DOES NOT have a 4–20 mA input.</b> Its sole process variable (PV) measurement input is a 3-wire Pt100 RTD input. Connecting 4–20 mA transmitter loops to input or output terminals will cause damage or operational failure. |
| LOOP WIRING CAUTION:<br>ACTIVE OUTPUT (DO NOT POWER)                     | The 4–20 mA analog output supplies its own internal excitation voltage (active source). Do NOT connect external DC power supplies in series with this loop; external voltage injection will burn out the onboard D/A converter stage.                                                                                                                                                                       |
| SENSOR REQUIREMENT:<br>3-WIRE Pt100 RTD ONLY                             | Requires a 3-wire Pt100 RTD sensor. Do not connect thermocouples (Type J/K) or uncompensated 2-wire RTD sensors where certified accuracy is required.                                                                                                                                                                                                                                                       |